

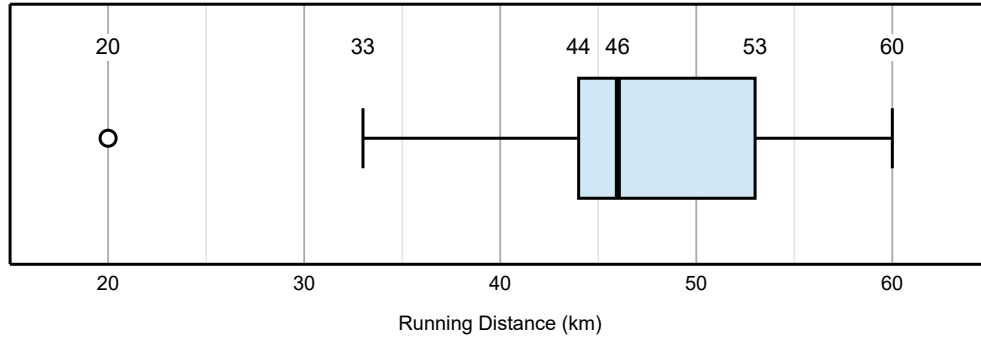
Introduction to Statistics - Quiz #1 (60 minutes)

September 17, 2026 (Thursday)

Section(교반): _____ Cadet Number(교번): _____ Name(성명): _____ Score: _____

- All solutions (except for the first problem) must include a detailed step-by-step explanation.
- If an answer has more than four decimal places, round to the fourth decimal place or write it as a fraction.

1. We randomly selected 17 students at a university and recorded each student's running distance (km) during the month of May. The boxplot below shows the results. Fill in each blank with the correct value. [30 points]



- (1) The median of the running distance is (_____).
- (2) The first quartile Q_1 of the running distance is (_____).
- (3) The outlier of the running distance is (_____).
- (4) The interquartile range (IQR) of the running distance is (_____).
- (5) About (_____)% of all the data values are between 44 and 53.

Solution: (1) 46 (2) 44 (3) 20 (4) 9 (5) 50

2. We surveyed 100 households in an apartment complex about the number of TVs X and the number of cars Y they have. The table below shows the number of households. Answer the following questions. [70 points]

$y \backslash x$	1	2	3	4
0	20	15	10	0
1	10	20	7	3
2	4	5	4	2

- (1) Fill in the table below with the joint probability distribution $f(x, y)$ of the random variables X and Y , and the marginal probability mass functions $f_X(x)$ and $f_Y(y)$. To find $f(x, y)$, divide each number in the table above by the total number of households, 100. For example, 20 households have 1 TV and 0 cars, so $f(1, 0) = 20/100 = 0.20$.

$y \backslash x$	1	2	3	4	$f_Y(y)$
0	0.20				
1					
2					
$f_X(x)$					

Solution: Divide each count by 100. Add each row to get $f_Y(y)$ and each column to get $f_X(x)$. From left to right: 0.20, 0.15, 0.10, 0.00, 0.45; 0.10, 0.20, 0.07, 0.03, 0.40; 0.04, 0.05, 0.04, 0.02, 0.15; 0.34, 0.40, 0.21, 0.05, 1.00

- (2) Calculate $E(XY)$, $E(X)$, and $E(Y)$.

Solution: Using the table in (1),

$$\begin{aligned}
 E(XY) &= \sum_{x=1}^4 \sum_{y=0}^2 xy f(x, y) = 1 \times 0 \times 0.2 + 2 \times 0 \times 0.15 + 3 \times 0 \times 0.1 + 4 \times 0 \times 0 \\
 &\quad + 1 \times 1 \times 0.1 + 2 \times 1 \times 0.2 + 3 \times 1 \times 0.07 + 4 \times 1 \times 0.03 \\
 &\quad + 1 \times 2 \times 0.04 + 2 \times 2 \times 0.05 + 3 \times 2 \times 0.04 + 4 \times 2 \times 0.02 = 1.51, \\
 E(X) &= \sum_{x=1}^4 x f_X(x) = 1 \times 0.34 + 2 \times 0.40 + 3 \times 0.21 + 4 \times 0.05 = 1.97, \\
 E(Y) &= \sum_{y=0}^2 y f_Y(y) = 0 \times 0.45 + 1 \times 0.40 + 2 \times 0.15 = 0.7.
 \end{aligned}$$

- (3) Calculate $\text{Cov}(X, Y)$.

Solution: Using the results of (2),

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 1.51 - 1.97 \times 0.7 = 1.51 - 1.379 = 0.131.$$

- (4) Determine whether the random variables X and Y are independent.

Solution: Check whether $f(x, y) = f_X(x) f_Y(y)$ holds for all (x, y) . For $(x, y) = (1, 0)$,
 $f(1, 0) = 0.2 \neq f_X(1) \times f_Y(0) = 0.34 \times 0.45 = 0.153$.

Therefore, X and Y are **not independent** (they are dependent).

- (5) One household is randomly selected. Let A be the event that this household has at least one car: $Y \geq 1$. Find $P(A)$.

Solution: By the complement rule,

$$P(A) = P(Y \geq 1) = 1 - P(Y = 0) = 1 - 0.45 = 0.55.$$

- (6) One household is randomly selected. Let B be the event that this household has three or more TVs: $X \geq 3$. Find $P(B | A)$.

Solution: By the definition of conditional probability,

$$P(B | A) = \frac{P(B \cap A)}{P(A)} = \frac{P(X \geq 3, Y \geq 1)}{P(Y \geq 1)} = \frac{0.07 + 0.03 + 0.04 + 0.02}{0.55} = \frac{0.16}{0.55} \approx 0.2909.$$